RAG Learning

# 6.1 Statistical Diagrams and Measures

OCR GCSE Mathematics J560 — Unit 6 — Papers 2 and 3 (Foundation and Higher)

## 6.1.1 Types of Data

| Type | Description | Examples |
|---|---|---|
| **Qualitative** | Non-numerical (categories) | Colour, gender, favourite subject |
| **Quantitative** | Numerical (measured or counted) | Height, number of goals, temperature |
| **Discrete** | Exact values only (usually counted) | Shoe size, number of students, goals scored |
| **Continuous** | Any value in a range (usually measured) | Height, weight, time, temperature |
| **Primary data** | Collected first-hand by the investigator | Surveys, experiments, observations |
| **Secondary data** | Collected by someone else, already exists | Census data, published reports, databases |

**Exam Tip:** Shoe size is a common trick: it looks continuous but is actually discrete (sizes are 4, 4.5, 5, … — not every value possible). Number of siblings is always discrete (you cannot have 2.3 siblings). Height and weight are always continuous.

## 6.1.2 Averages and Range

The three averages and range are:

- **Mean**: sum of all values ÷ number of values. Notation: $\bar{x} = \frac{\sum x}{n}$
- **Median**: middle value when data is arranged in order. For $n$ values, position is $\frac{n+1}{2}$
- **Mode**: most frequent value (there can be more than one, or none)
- **Range**: largest value — smallest value (measures *spread*, not average)

## Comparing the Averages

| Average | Advantages | Disadvantages | Best used when |
|---|---|---|---|
| **Mean** | Uses all data values; most precise | Distorted by outliers | Data is roughly symmetrical, no extreme values |
| **Median** | Not affected by outliers | Ignores most values | Data is skewed or has outliers (e.g. salaries) |
| **Mode** | Easy to find; only average for qualitative data | May not exist or may not be representative | Finding most common category (e.g. most popular shoe size) |

## Averages from a Frequency Table

**Worked Example:** The table shows the number of books read by 20 students last month.

| Books read ($x$) | Frequency ($f$) | $f \times x$ |
|---|---|---|
| 0 | 3 | 0 |
| 1 | 5 | 5 |
| 2 | 7 | 14 |

| | | |
|---|---|---|
| 3 | 4 | 12 |
| 4 | 1 | 4 |
| **Total** | **20** | **35** |

**Mean** $= \dfrac{\sum fx}{\sum f} = \dfrac{35}{20} = 1.75$ books

**Mode** $= 2$ (highest frequency = 7)

**Median**: $\dfrac{20+1}{2} = 10.5$th position → average of 10th and 11th values.
Cumulative frequencies: 3, 8, 15, ... The 10th and 11th values are both in the group with $x = 2$.
**Median = 2**

## Averages from a Grouped Frequency Table

When data is grouped, we cannot find exact values. We use the **midpoint** of each class interval.

**Worked Example:**

| Height $h$ (cm) | Frequency | Midpoint | $f \times \text{midpoint}$ |
|---|---|---|---|
| $150 \leq h < 160$ | 4 | 155 | 620 |
| $160 \leq h < 170$ | 11 | 165 | 1815 |
| $170 \leq h < 180$ | 9 | 175 | 1575 |
| $180 \leq h < 190$ | 6 | 185 | 1110 |
| **Total** | **30** | | **5120** |

**Estimated mean** $= \dfrac{5120}{30} \approx 170.7$ cm

**Modal class** $= 160 \leq h < 170$ (highest frequency)

**Median class**: $\frac{30}{2} = 15$th value → cumulative frequencies: 4, 15, 24... → median lies in $160 \leq h < 170$

**Exam Tip:** For grouped data, you can only estimate the mean (using midpoints). Always state it is an estimate. The modal class is the class with the highest frequency — do not just write the midpoint.

## 6.1.3 Bar Charts, Pictograms and Pie Charts

### Pie Charts

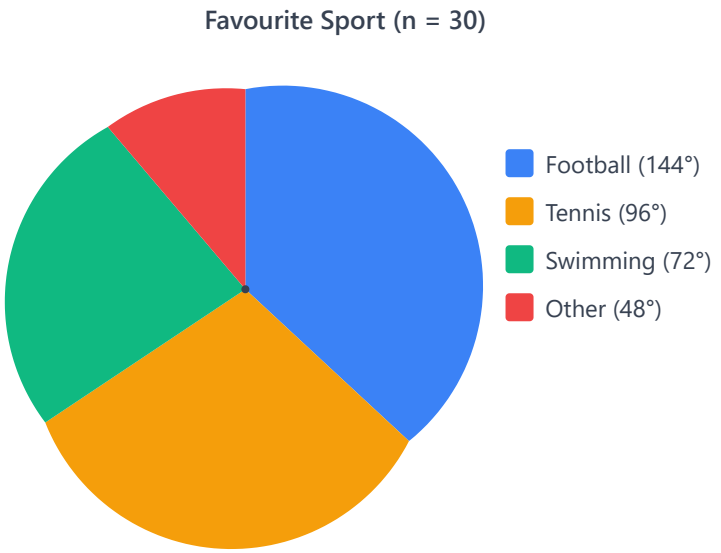
In a pie chart, each sector represents a category. The **angle** of each sector is:

$$\text{Angle} = \frac{\text{frequency}}{\text{total frequency}} \times 360^\circ$$

**Worked Example:** 30 students were asked their favourite sport. Results: Football 12, Tennis 8, Swimming 6, Other 4.

| Sport | Frequency | Angle calculation | Angle |
|---|---|---|---|
| Football | 12 | $\frac{12}{30} \times 360$ | 144° |
| Tennis | 8 | $\frac{8}{30} \times 360$ | 96° |
| Swimming | 6 | $\frac{6}{30} \times 360$ | 72° |
| Other | 4 | $\frac{4}{30} \times 360$ | 48° |

| | | | |
|---|---|---|---|
| **Total** | **30** | | **360°** |




**Exam Tip:** When drawing a pie chart, always check your angles add to 360°. When reading a pie chart to find a frequency, rearrange the formula: frequency = (angle / 360) × total.

## 6.1.4 Frequency Diagrams and Polygons

A **frequency polygon** is drawn for grouped data by:

1. Finding the **midpoint** of each class interval.
2. Plotting (midpoint, frequency) for each class.
3. Joining the points with straight lines.
4. Extending the line to the midpoints of the empty classes at each end (touching the x-axis).

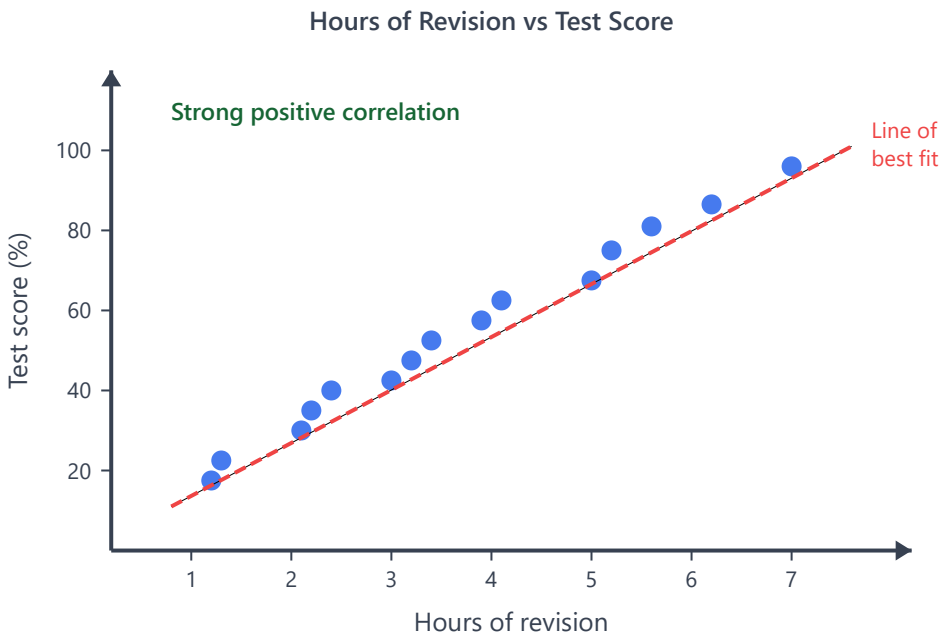
Frequency polygons are particularly useful for comparing two distributions on the same diagram.

**Exam Tip:** A frequency polygon is NOT a bar chart. Do not shade under the lines. Plot points at midpoints, not at the boundaries of the class intervals.

# 6.1.5 Scatter Diagrams and Correlation

A **scatter diagram** shows the relationship between two variables. Each point represents one data item (its two measurements).

- **Positive correlation**: as one variable increases, so does the other.
- **Negative correlation**: as one variable increases, the other decreases.
- **No correlation**: no clear pattern between the variables.
- Correlation can be **strong** (points close to a line) or **weak** (points more scattered).




## Line of Best Fit

A **line of best fit** (or trend line) is a straight line that best represents the data on a scatter diagram. It passes through the mean point $(\bar{x}, \bar{y})$ and has roughly equal numbers of points above and below it.

## Interpolation vs Extrapolation

| Interpolation | Extrapolation |
|---|---|
| Using the line of best fit **within** the data range | Using the line of best fit **outside** the data range |
| Generally reliable | Less reliable — the pattern may not continue |

### Correlation vs Causation

**Correlation** does not imply **causation**. Just because two variables are correlated does not mean one causes the other — there may be a third (confounding) variable, or the correlation may be coincidental.

**Exam Tip:** In exam questions about scatter diagrams: (1) describe the correlation (strong/weak, positive/negative/none), (2) use the line of best fit to make predictions only within the data range, (3) if asked whether correlation implies causation, the answer is always no — explain why.

## 6.1.6 Time Series and Line Graphs

A **time series** is a set of data recorded at regular time intervals. Points are joined with straight lines to show trends over time.

When reading a time series graph:

- Identify the overall **trend** (increasing, decreasing, constant).
- Identify any **seasonal variation** (regular up-down pattern).

**Higher Tier Only:** **Moving Averages** smooth out seasonal variation to reveal the underlying trend.

For a **4-point moving average**, average each group of 4 consecutive values:

$$\text{MA}_1 = \frac{Q1 + Q2 + Q3 + Q4}{4}, \quad \text{MA}_2 = \frac{Q2 + Q3 + Q4 + Q5}{4}, \quad \ldots$$

Plot each moving average at the *midpoint* of its group of 4 values.

**Worked Example:** Quarterly sales (£000s): Q1=12, Q2=18, Q3=22, Q4=14, Q1=15, Q2=20

**4-point moving averages:**

MA1 $= \dfrac{12 + 18 + 22 + 14}{4} = \dfrac{66}{4} = 16.5$

MA2 $= \dfrac{18 + 22 + 14 + 15}{4} = \dfrac{69}{4} = 17.25$

MA3 $= \dfrac{22 + 14 + 15 + 20}{4} = \dfrac{71}{4} = 17.75$

The moving averages are increasing from 16.5 to 17.75, suggesting an upward trend.

**Exam Tip:** Moving averages are plotted at the mid-time point between the first and last data points in each group. For a 4-point MA, plot between the 2nd and 3rd point of each group of 4. The trend line is drawn through the moving average points.

## 6.1.7 Histograms

A **histogram** is used for continuous grouped data. Unlike a bar chart, the **area** of each bar represents the frequency — not the height. This is necessary because class widths may vary.

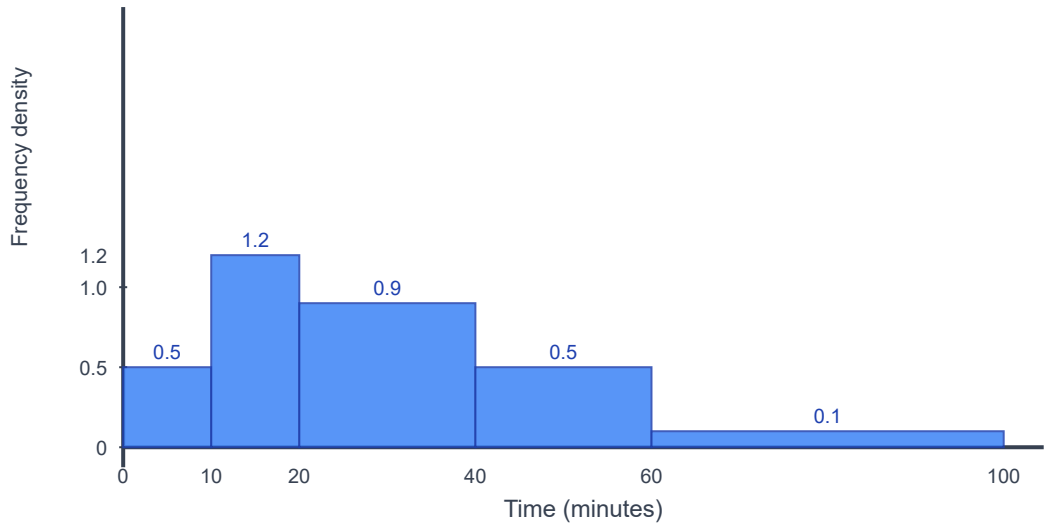
$$\text{Frequency density} = \frac{\text{Frequency}}{\text{Class width}}$$

$$\text{Frequency} = \text{Frequency density} \times \text{Class width}$$

**Worked Example:** The table shows the times (in minutes) students spent on homework. Draw a histogram.

| Time $t$ (min) | Frequency | Class width | Freq. density |
|---|---|---|---|
| $0 \leq t < 10$ | 5 | 10 | 0.5 |
| $10 \leq t < 20$ | 12 | 10 | 1.2 |
| $20 \leq t < 40$ | 18 | 20 | 0.9 |
| $40 \leq t < 60$ | 10 | 20 | 0.5 |
| $60 \leq t < 100$ | 4 | 40 | 0.1 |

Note: the bar for $20 \leq t < 40$ is twice as wide but has frequency density 0.9, giving area $20 \times 0.9 = 18$ — correctly representing a frequency of 18.




Histogram of homework times. Bar area = frequency. Note the wide bars for 20–40 and 60–100 min.

**Exam Tip:** The y-axis of a histogram must be labelled "Frequency density", not "Frequency". If a question gives a partly-drawn histogram and asks you to find a missing frequency, calculate: frequency = frequency density × class width using the scale on the y-axis.

## 6.1.8 Cumulative Frequency and Box Plots

### Cumulative Frequency

**Cumulative frequency** is a running total of frequencies up to the end of each class. Plot cumulative frequency against the **upper class boundary** to draw a cumulative frequency curve (ogive).

**Worked Example:** Heights of 60 students:

| Height $h$ (cm) | Frequency | Cumulative frequency |
|---|---|---|
| $150 \leq h < 160$ | 8 | 8 |

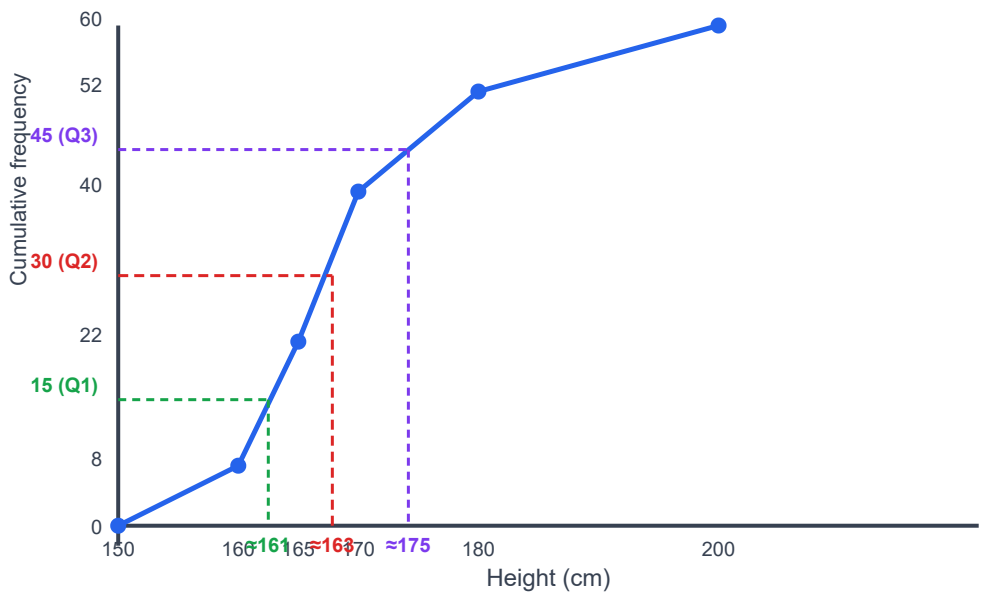
| | | |
|---|---|---|
| $160 \leq h < 165$ | 14 | 22 |
| $165 \leq h < 170$ | 18 | 40 |
| $170 \leq h < 180$ | 12 | 52 |
| $180 \leq h < 200$ | 8 | 60 |

Plot points at: (150, 0), (160, 8), (165, 22), (170, 40), (180, 52), (200, 60) — join with a smooth curve.

## Reading from a Cumulative Frequency Curve

From a cumulative frequency curve for $n$ data values, read off:

- **Median**: the value at cumulative frequency $\frac{n}{2}$
- **Lower quartile (Q1)**: the value at cumulative frequency $\frac{n}{4}$
- **Upper quartile (Q3)**: the value at cumulative frequency $\frac{3n}{4}$
- **Interquartile range (IQR)**: $\mathrm{IQR} = Q_3 - Q_1$
- **Percentiles**: $p$th percentile is at cumulative frequency $\frac{pn}{100}$

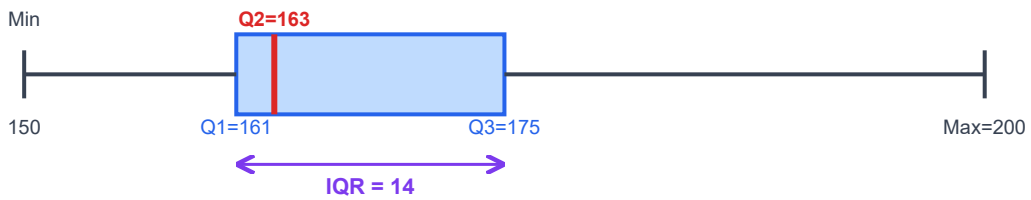



Cumulative frequency curve. Red dashed: median (Q2). Green: lower quartile (Q1). Purple: upper quartile (Q3).

## Box Plots (Box-and-Whisker Diagrams)

A **box plot** shows the distribution of data using five key values:

- Minimum value (or lower fence if outliers are excluded)
- Lower quartile Q1
- Median Q2
- Upper quartile Q3
- Maximum value (or upper fence if outliers are excluded)




Box plot for heights data. The box spans Q1 to Q3 (the IQR). The red line is the median. Whiskers extend to minimum and maximum.

**Exam Tip:** When comparing two box plots, comment on: (1) the **median** (typical value — which is higher?), (2) the **IQR or range** (spread — which is more consistent/variable?), (3) the **skewness** (if the median is closer to Q1, the data is positively skewed; closer to Q3, negatively skewed). Always use the context of the question in your comparison.

# 6.1.9 Outliers and Interquartile Range

An **outlier** is a data value that is unusually far from the rest of the data. The standard definition used in GCSE/A-Level:

$$\text{Outlier if } x < Q_1 - 1.5 \times \text{IQR} \quad \text{or} \quad x > Q_3 + 1.5 \times \text{IQR}$$

These limits are called the **lower fence** and **upper fence**.

**Worked Example:** Data set: 12, 15, 18, 21, 23, 25, 27, 30, 55.

**Step 1:** Arrange in order (already done). $n = 9$.
**Q1** = 18 (lower half: 12, 15, 18, 21; median = 16.5... actually Q1 = median of lower half = 15+18)/2=16.5)
Let's compute properly: Q1 = $\frac{15+18}{2} = 16.5$, Q2 = 23, Q3 = $\frac{27+30}{2} = 28.5$
**IQR** = 28.5 − 16.5 = 12
**Lower fence** = 16.5 − 1.5(12) = 16.5 − 18 = −1.5
**Upper fence** = 28.5 + 1.5(12) = 28.5 + 18 = 46.5
**Outlier check:** 55 > 46.5, so **55 is an outlier**. All other values are between −1.5 and 46.5.

**Exam Tip:** On a box plot, outliers are shown as individual crosses (×) or dots beyond the whiskers. The whiskers then extend only to the most extreme values that are *not* outliers (the fences). Always state the fence values when asked to identify outliers.

## 6.1.10 Standard Deviation

**Standard deviation** measures the average spread of data around the mean. A larger standard deviation means data is more spread out.

$$\sigma = \sqrt{\frac{\sum(x-\bar{x})^2}{n}} = \sqrt{\frac{\sum x^2}{n} - \bar{x}^2}$$

The second formula (using $\sum x^2$) is usually easier to compute.

**Worked Example:** Find the mean and standard deviation of: 3, 5, 7, 8, 12.

$$\bar{x} = \frac{3+5+7+8+12}{5} = \frac{35}{5} = 7$$

$$\sum x^2 = 9 + 25 + 49 + 64 + 144 = 291$$

$$\sigma = \sqrt{\frac{291}{5} - 7^2} = \sqrt{58.2 - 49} = \sqrt{9.2} \approx 3.03$$

## Standard Deviation from a Frequency Table

For data in a frequency table with values $x_i$ and frequencies $f_i$:

$$\bar{x} = \frac{\sum f_i x_i}{\sum f_i} \qquad \sigma = \sqrt{\frac{\sum f_i x_i^2}{\sum f_i} - \bar{x}^2}$$

For **grouped data**, use the midpoint of each class in place of $x_i$.

**Worked Example:** Find the standard deviation for the grouped height data (using midpoints):

| Class (cm) | Midpoint $x$ | Freq $f$ | $fx$ | $fx^2$ |
|---|---|---|---|---|
| $150 \leq h < 160$ | 155 | 4 | 620 | 96,100 |
| $160 \leq h < 170$ | 165 | 11 | 1,815 | 299,475 |
| $170 \leq h < 180$ | 175 | 9 | 1,575 | 275,625 |
| $180 \leq h < 190$ | 185 | 6 | 1,110 | 205,350 |
| **Total** | | **30** | **5,120** | **876,550** |

$$\bar{x} = \frac{5120}{30} \approx 170.67 \text{ cm}$$

$$\sigma = \sqrt{\frac{876{,}550}{30} - 170.67^2} = \sqrt{29{,}218.3 - 29{,}128.5} = \sqrt{89.8} \approx 9.48 \text{ cm}$$

This is an *estimate* because we used midpoints.

**Exam Tip:** Standard deviation is always non-negative. A standard deviation of 0 means all values are identical. For grouped data, always use midpoints and call the answer an estimate. On a calculator, use the $\sigma_n$ function (not $\sigma_{n-1}$, which is the sample standard deviation) unless told otherwise.

## Comparing Distributions

| Measure | What it tells you | Use when comparing |
|---|---|---|
| Mean | Average (centre) | Which group has a higher typical value? |
| Median | Middle value (resistant to outliers) | When data is skewed or has outliers |
| Range | Total spread | Quick comparison; affected by outliers |
| IQR | Spread of middle 50% | More robust measure of spread; used with box plots |
| Standard deviation | Average distance from mean | Most detailed measure of spread |

**Exam Tip:** Comparison questions always need two statements: one about **average** (median or mean) and one about **spread** (IQR or standard deviation). Both must use the data context — e.g. "Group A had a higher median score (45 vs 38), suggesting they performed better on average. Group A also had a smaller IQR (10 vs 18), suggesting their scores were more consistent."

© RAG Learning · Not for redistribution